

RESEARCH STATEMENT

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My research focuses primarily on **Topological Data Analysis (TDA)**. Based on the principle that data possesses an intrinsic shape, TDA seeks to extract meaningful information encoded in the topological and geometrical features of its shape. My research agenda is organized around three directions: applying topological methods to real-world data, developing efficient computational implementations of these methods, and investigating their theoretical foundations through category theory to provide more general frameworks. This latter direction has unexpectedly led to connections with **combinatorics**, particularly in the theoretical and applied aspects of order theory, which has become an emerging focus of my research.

Past research: I worked with Professor Jose Perea and Francisco Gómez to reveal the brain network dynamics during the emotion of suspense using Mapper (§3). Mapper is a topological method introduced in 2007 by Singh, Memoli and Carlsson [1] that simplifies high dimensional data into simplicial complexes while preserving structural information. In particular, we obtained a graph that summarizes the dynamical connectivity of the brain while experiencing suspense. Using this method, we demonstrated that the functional connections inside and between the large-scale networks of the brain change depending on the level of suspense.

Present research: Joint with my advisor Elizabeth Munch, we bound the interleaving distance between generalized persistence modules on concrete categories (§1). The interleaving distance is one of the most used metrics to quantify the dissimilarity between geometric-topological structures that serve as fundamental descriptors in TDA like Mapper graphs, Reeb graphs and persistence modules. While the interleaving distance is widely used in theory, the ability to be used in practice is limited by its computational complexity in the vast majority of settings. Motivated by this limitation we generalized the ideas introduced by Chambers et al. [2] in the context of Mapper graphs. Explicitly, we defined a loss function that provides a framework for computing upper bounds on the interleaving distance for any descriptor. Then, we demonstrated, under reasonable settings, that our general framework enables the existence of a polynomial-time computable upper bound on the interleaving distance between multiparameter persistence modules, one of the primary algebraic structures used in TDA [3].

In this general framework, one can consider a partially ordered set (poset) a category. Using the notion of interleaving distance, I introduced a new family of distances over partially ordered sets (§2). Posets have become increasingly relevant in a variety of applications involving data that is naturally endowed with an ordered structure. Thus, many problems require the notion of a distance that reflects the underlying order relations. I established a new family of extended metrics posets that do not require additional structure creating the possibility of the study of a broader range of data structures. Additionally, the metrics are an invariant capable of reconstructing the underlying poset, thereby addressing a general reconstruction problem.

Future research: My short-term objective is to develop and implement computational tools through guided research with undergraduate and graduate students. I plan to pursue this objective by formulating concrete and well-defined projects from my research allowing students to make meaningful contributions.

First, in the context of the interleaving distance, my next objective is bounding the distance between multiparameter persistence modules in polynomial time (§1). The problem naturally reduces to computations with finite-dimensional vector spaces, requiring primarily a solid background in linear algebra and programming. This approach is not without precedent; for example, the mentioned background provides sufficient preparation for introducing computational homological algebra to undergraduate students [4].

Second, in the context of the new family of distances over posets, I aim to create an algorithm that can recover the order of poset using solely distance values and optimize the existing algorithms for calculating the distances (§2). These projects will strengthen the applicability of the new metrics while providing natural opportunities for undergraduate research. Since order theory can be approached with relatively few prerequisites, including basic set theory and mathematical logic, students can engage with these questions at an early stage of their mathematical development.

My medium-term objective is to develop interdisciplinary collaborations through the application of combinatorics and TDA to real-world data. On one hand, I aim to investigate datasets that naturally carry an ordered structure using the proposed family of poset metrics (§2). On the other hand, I seek to use Mapper and its variations to investigate the “shape of relations” (§3). This topological technique has already been successfully applied across a variety of disciplines [5]. Together, these directions will allow me to extend my research toward problems motivated by real data and collaboration with researchers from other fields.

In the following, I provide a more detailed discussion of each of my research projects. I elaborate on the background and motivations of each project, show the main results we obtained and conclude by outlining the future directions I intend to pursue.

1. BOUNDING THE INTERLEAVING DISTANCE IN POLYNOMIAL TIME

Given a finite dataset, its topological properties can be encoded by a generalized persistence module, which is defined as a functor F from a partial ordered set P to any category C [6]. This categorical viewpoint provides a unified framework for multiple topological descriptors. A fundamental requirement for any descriptor is stability: small changes in the input should produce only small changes in the corresponding functors, thus, it is natural to desire a notion of proximity between generalized persistence modules. The most widely used metric is the interleaving distance, which has been used in many stability results in TDA. However, computing the interleaving distance has been shown to be NP-hard for several important classes of persistence modules [7]. This research addresses this computational challenge by deriving upper bounds for the interleaving distance from the categorical perspective of generalized persistence modules. This framework is then used to

develop polynomial-time computable approximations to the interleaving distance.

Let F and G be two generalized persistence modules. A pair of natural transformations (φ, ψ) is an ε -interleaving between F and G if it approximates the notion of isomorphism between F and G by an ε -shift. In my work, we consider (φ, ψ) a pair of ε -assignments, a weakened notion of ε -interleaving, and then we measure by $L(\varphi, \psi)$ how far the assignment is from being an interleaving. We show that this loss provides an upper bound for the interleaving distance between F and G , namely

$$d_I(F, G) \leq \varepsilon + L(\varphi, \psi)$$

In this way, the computational challenge of the is reduced to the computation of an associated loss function over the space of ε -assignments. We further show that this loss can be efficiently computed and optimized for multiparameter persistence modules. Explicitly, when the indexing poset is given by $P = \mathbb{R}^k$, the finite cartesian product of the real line, and the target category is $C = \text{Vec}_{\mathbb{F}}$, the category of vector spaces over a field $\mathbb{F}$.

Having proved that the loss function admits a polynomial-time computation, my next objective is to leverage this result to a practical polynomial-time algorithm for approximating the interleaving distance between multiparameter persistence modules. Beyond algorithm design, I aim to establish theoretical guarantees on the tightness of the approximation and identify invariants of persistence modules that further simplify the computation. Through this work, I seek to advance the computational tractability of multiparameter persistence, contributing to the broader effort of making this powerful descriptor more accessible for both theoretical research and practical applications [8–11].

Additionally, our general framework applies to any generalized persistence module including merge trees, Reeb graphs, sheaves and cosheaves. Consequently, the proposed loss yields an upper bound on the interleaving distance between such structures. In the future, I seek to apply this framework to develop practical methods for bounding and approximating these distances. The development of algorithms to compute this loss is inherently context dependent, relying on the specific properties of each structure. For example, the existence of basis elements when the indexing poset P consists of collections of open sets [2] or P has properties like modularity and lattice ordering. For instance, sheaves and cosheaves are naturally indexed by distributive lattices, potentially providing a structure to develop efficient algorithms.

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2. A NEW FAMILY OF DISTANCES OVER PARTIALLY ORDERED SETS

Partially ordered sets (posets) and order theory are playing an increasingly important role in applications across multiple fields [12–15]. Their growing relevance stems from the fact that many real-world datasets are naturally endowed with an ordered structure. Thus, a fundamental problem

is to quantify the dissimilarity between data points by defining meaningful notions of distance between elements of a poset. Several metrics have been proposed for this purpose, however, many rely on additional structural assumptions such as modularity, lattice structure or boundedness, that limits their applicability to more general ordered data.

My research addresses this limitation by introducing two new families of extended metrics defined over posets, called the k -climber and k -diver metrics for k natural. These metrics are a type of shortest-path metric that depend on the length and number of alternations of a path. Consequently, they can be defined on any path-connected poset without requiring additional structural assumptions. Moreover, I prove that the 1-climber and 1-diver distance characterize most discrete posets up to isomorphism and modular posets up to duality, thereby establishing new reconstruction results from metric information. In addition to these theoretical results, I have implemented several functions in SageMath for computing the proposed metrics, providing a computational foundation for further study and applications.

This project gives rise to several avenues for future research, which I intend to explore in collaboration with colleagues and through the mentoring of graduate or undergraduate students. First, it is known that if the shortest-fence and shortest-path distances of a finite poset are given, then the underlying order and its dual can be recovered by a computable algorithm [16]. Since I have shown that finite posets are likewise characterized by their 1-climber and 1-diver metrics, it is natural to conjecture that an analogous reconstruction algorithm can be developed using these distance values. Such an algorithm could provide a new method for recovering a poset directly from its metric information.

A second direction concerns the computational aspects of these metrics. An important open problem is determining the algorithmic complexity of computing the k -climber and k -diver distances. Future work will focus on establishing complexity bounds, designing optimized algorithms and identifying classes of posets for which these computations can be performed efficiently. The development of better implementations would strengthen the applicability of these metrics.

Finally, returning to the original motivation for this project, I aim to understand which structural and hierarchical features of ordered data these metrics capture and how they can be exploited in the analysis of real-world datasets. This direction has the potential to generate interdisciplinary collaborations with researchers in fields where data naturally carries an ordered structure, extending the applicability of these methods beyond mathematics.

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3. MAPPER AND THE SHAPE OF RELATIONS

Advances in technology have enabled researchers across various disciplines to generate vast amounts of interesting, yet complex and high dimensional data. The challenge now translates on how to

extract the valuable information inherent in such datasets and visualize its hidden patterns. The Mapper algorithm addresses these challenges by simplifying high-dimensional data equipped with a dissimilarity measure into a simplicial complex whose edges represent relations between data points thereby preserving structural information. This graphical representation provides an effective way to encode relationships within data, making Mapper particularly valuable for data exploration and visualization.

During my master’s research, I applied Mapper to reveal large-scale dynamic organization of the brain. This experience gave me firsthand insight into both its limitations and its potential. On the one hand, a major limitation of Mapper is that its output depends on the selection of multiple parameters making successful results dependent on a parameter fine tuning process. Over the years, the community has developed variants of Mapper including Ball Mapper, F-Mapper, G-Mapper among others to overcome this limitation [5]. Looking forward, I would like to explore the application of these variants with the goal of understanding what additional questions they can address and what new information they can extract from data.

On the other hand, the Mapper algorithm has been successfully applied across a variety of fields where high-dimensional data is prevalent like neuroscience [17], economics [18], and Large language models (LLMs) [19]. As heterogeneous as these fields are, the potential of Mapper lies in its ability to reveal hidden relationships between data points. Going forward, I am interested in pursuing interdisciplinary research in which datasets generated in other fields provide opportunities to test and adapt Mapper variants. Moreover, I would collaborate with colleagues and students to generate new insights from these datasets, turning the research into collaborative projects with practical outcomes.

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